

# The Boot Process

## Kernel Ring Buffer

kernel log messages

in-memory data structure

captures:

- boot messages
- event logging
- errors and warnings

### **dmesg**

print or control kernel ring buffer

```
dmesg [options]
```

## The Boot Process

Steps when powering on a computer

1. system powered on
2. CPU runs firmware (BIOS or EFI)
3. firmware looks for *Boot Loader*
4. Boot Loader loads the kernel or *chainloads* to another boot loader
5. kernel initializes devices, mounts root partition and runs initial program for your system